H-1B Visas & Worker Outcomes: Updates

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Overview

Research Question: do workers at H-1B winning firms have better outcomes than workers at H-1B losing firms?

- Key contributions: use new dataset (Bloomberg H-1B data merged with LinkedIn) to (1) answer crowd-out question better and (2) dig into new outcomes
- Empirical strategy: compare hires, promotions, exits at firms who win H-1B lottery vs. lose H-1B lottery
- Progress so far:
  1. Descriptives and summary stats for both datasets
  2. Merging firms—65% merged currently, can almost certainly do better but tabling for now
  3. Extremely preliminary results on crowd-out

Outline

1. Overview
2. [SKIP] Background and Data
3. H-1B Data: Descriptives and Challenges
4. Merging with Revelio Data
5. Initial Results
6. Next Steps

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H-1B Visas: Policy Overview

- Nonimmigrant visa for workers in 'specialty occupations'
  - "A specialty occupation is one that requires the application of a body of **highly specialized knowledge** and the attainment of at least a bachelor's degree or its equivalent. The intent of the H-1B provisions is to help employers who **cannot otherwise obtain needed business skills and abilities from the U.S. workforce...**"
- Employment-based visa: employer must apply on worker's behalf
  - 'H-1B portability' between employers introduced in 2000, but lots of friction remains
  - 'Dual-intent' visa: employer can sponsor green card for worker
- Generally a three-year visa with a three-year extension
- Capped at 65,000 normal visas annually and 20,000 advanced-degree exempt (ADE) visas for U.S. advanced degree holders (some employers exempt from cap)

H-1B Filing Process

- To hire H-1B worker, firm first required to file Labor Condition Application (LCA) with DOL attesting that the job will not replace/harm incumbent workers
  - Not tied to a specific employee, must be filed no more than six months before start date
  - Includes information about job type (occupation code), start and end dates, work location, and salary
- Prior to FY2021, firms required to fill out full paperwork for H-1B (including LCA and filing fees of $\sim$\$500 per applicant). Only forms of lottery winners processed, rest thrown out.
- Starting FY2021, firms only required to fill out short form and \$10 filing fee to enter lottery, then only lottery winners required to fill out full paperwork.
  - Bloomberg investigation—streamlined process caused fraud to skyrocket (multiple submissions of same employee from different companies)

Sources of Variation: Literature and Context

- Quota of 65,000 established in 1990, increased in 1999 then again in 2001, reset to 65,000 + 20,000 ADE in 2004
  - Older literature using these aggregate increases as source of variation [Kerr and Lincoln 2010, Peri, Shih and Sparber 2015, Bound, Khanna and Morales 2017]
- Initially, visas allocated on a first-come, first-served basis until cap reached
- In FY2006-7, USCIS ran unannounced pilot lotteries for applicants on day that cap was reached
  - Small sample of 3k firms **but** includes data on lottery losers [Doran, Gelber and Isen 2022 (DGI)]
- In FY2008 (and most future years) the cap was reached on the first day of the filing period, so *all* visas were distributed via lottery
  - All firms **but** no data on lottery losers [Dimmock, Huang and Weisbenner 2022 (DHW); Mahajan et al. 2024 (MMSCB)]
  - Instead, DHW and MMSCB use LCAs as imperfect proxy for H-1B applications

Data

- **H-1B Data**
  - Bloomberg FOIA request [FY2021-2024]: microdata on *all H-1B applications*, i.e. both lottery winners and losers
  - Previous FOIA request [FY2017-2024]: microdata only on H-1B lottery winners
  - US DOL [FY2009-2024]: microdata on LCAs (used in literature as proxy for employer H-1B demand)
  - USCIS Employer Hub [FY2009-2024]: employer-level data on H-1B lottery winners (missing person-level characteristics)
- **Revelio LinkedIn Data**
  - Position-level data from universe of LinkedIn profiles
- **Merging by Employer** (Currently 60% of H-1B Employers FY2021-4 linked)
  - Messy! Limited info with lots of missings, need to merge on name and location only
  - Would love to link at the individual level, but limited info on both ends (future todo)

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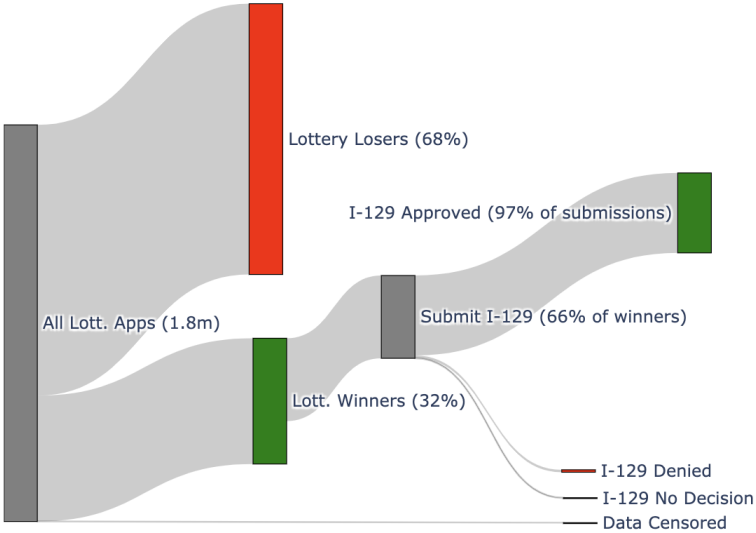
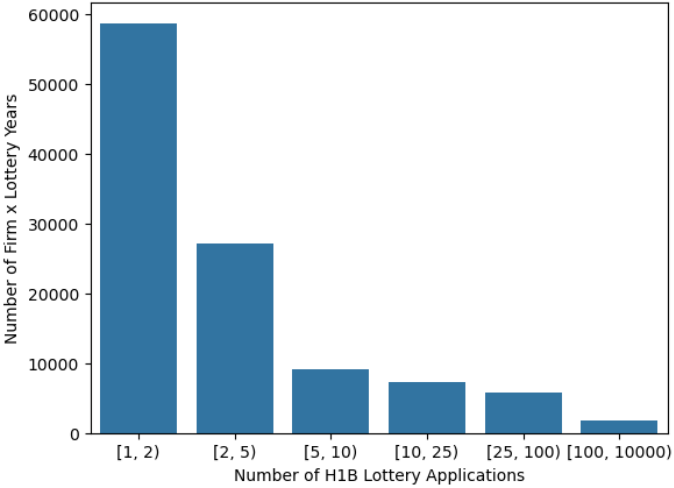
Bloomberg Data: Summary

- Covers four years of lotteries: FY2021 (March 2020 Lottery)-FY2024 (March 2023 Lottery)
- Includes lottery application data and lottery status for all H-1B applicants *subject to the lottery*
  - Includes both regular and Advanced Degree Exempt (ADE) applicants (subject to different lotteries) **but does not identify the group into which the applicant falls**
  - Does not include any information on cap-exempt employers (non-profits, higher ed, etc.), applications for change in employer or visa extension
  - Contrast to previous FOIA data which has no information on lottery losers
- Additionally, includes information from I-129 (H-1B Visa Application Form) for lottery applicants who won **and** submitted form (66% of winners)

Sample Sizes: All Years of Bloomberg Data



Lottery App Dist. (Firm x Year Counts FY2021-2024)



Raw Histogram

Company Sample Selection

- Very long right tail
  - Top 20 employers submit 20% of applications
- Well-documented issue with outsourcing/IT staffing companies flooding H-1B lottery
- Want a sample of small companies most likely to see effect of H-1B win on workers and not dependent on H-1Bs
- **Preferred Sample:** Non-staffing companies with fewer than 50 H-1B lottery applications [138k emp x lottery years (80% of total employers; 24% of total applications)]
  - Staffing companies IDed as those with at least 50% of applications marked as duplicate entries (can probably do better)
  - $<$ 4% of employers submit 50 or more applications
  - Preferred sample [PS] has much higher win rate (36% vs. 30% for out of sample [OOS]), likely from higher share of ADE applicants
  - Applicants from OOS firms overwhelmingly from India (85% vs. 51% in PS)

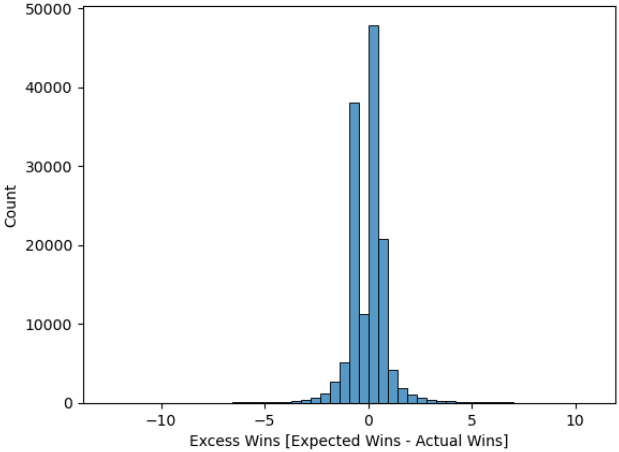
H-1B Lottery Winners and Losers (Preferred Sample)

	Lottery Winners	Lottery Losers
% Born in India	52.0%	49.8%
% Born in China	18.4%	14.0%
% Female	37.4%	34.9%
Avg Age at Application	30.6	31.8
N	156,560	273,052

- All differences significant at 99% level
- Likely represent differences in winning probability between normal applicants and ADE applicants—this is most concerning empirical issue currently

▸ Age Distribution

Dist. of Excess Wins (Firm j x Year t Counts FY2021-2024)



$$\text{Excess Wins}_{jt} = \text{Lottery Apps}_{jt} \times \text{Total Win Rate}_t - \text{Actual Wins}_{jt}$$

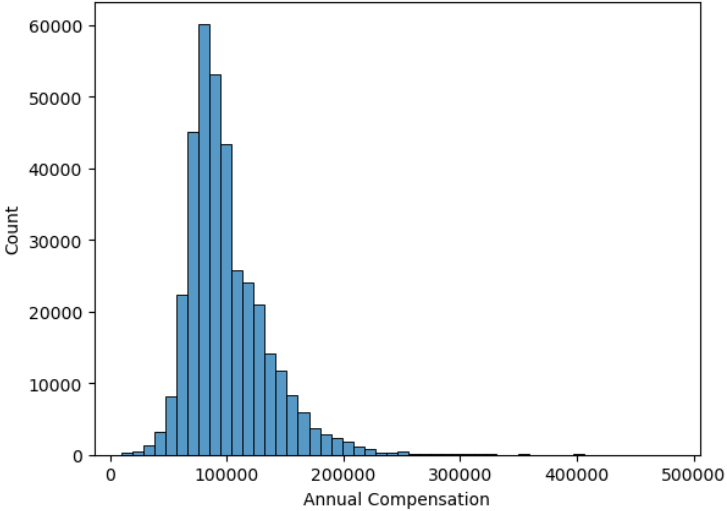
▸ Distribution of Win Rates

Complier Characteristics

Much more information on those who submit I-129s (win lottery and subsequently apply for visa):

- 42.4% new admit, 57.4% change status
  - Lots of missings in current visa field, but of those who report vast majority on F1 (65%), otherwise L (multinational corp), H4 (spouse of H1B) TN (Canada and Mexico)
- 76% applying regularly, 24% applying with ADE
  - Approx. 100% of ADE applicants have more than a bachelor's degree
  - Even of regular applicants, 45% have a master's or doctorate
- 70% computer/EE related occupations
- 47% at firms with NAICS code 5415 (Computer Systems Design and Related Services)

Dist. of Compensation Paid



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Merging Strategy

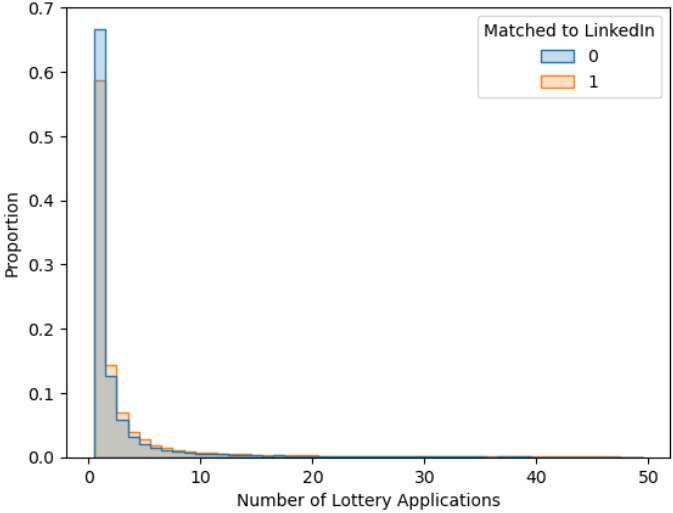
Current Merge (66% of Preferred Sample):

- Keep at company x lottery year level (inconsistent identifiers across years)
- Initial (tokenized) matching on company name
- Coarse filter on matches based on name similarity, name rarity, employer/worksite state, NAICS code (if reported)
- Matched firms submit more applications and lose slightly more on average than unmatched firms

Future Merge:

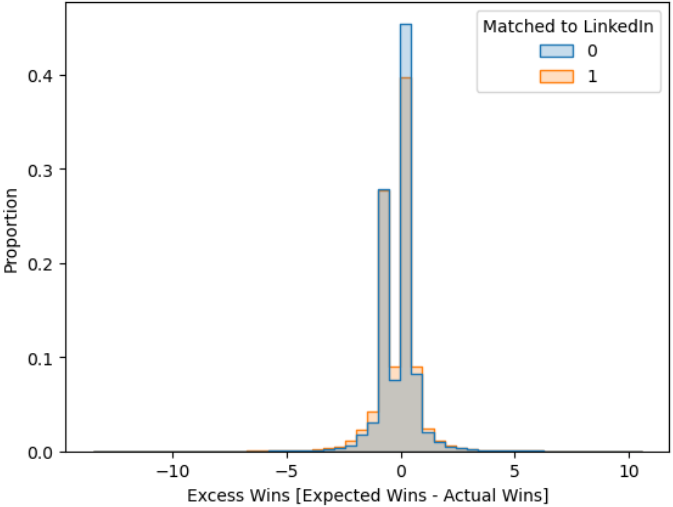
- Include more detailed info on location
- Probabilistic linking
- Better de-duplication of both datasets

Merging Balance: Applications by Merge



Matched Mean is 3.3, Unmatched Mean is 2.7***

Merging Balance: Excess Wins by Merge



Matched Mean is -0.07, Unmatched Mean is -0.02^{***}

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Ideal Empirical Strategy and Outcomes of Interest

Following DGI, for firm j in year t who applied for an H-1B visa in fiscal year T, estimate:

$$Y_{jtT} = \beta_0 + \beta_1 U_{jT} + \varepsilon_{jtT} \tag{1}$$

where U_{jT} represents 'excess' lottery wins above chance (i.e. number of lottery wins net of number of applications $\times$ aggregate win rate).

I am interested in the following outcomes:

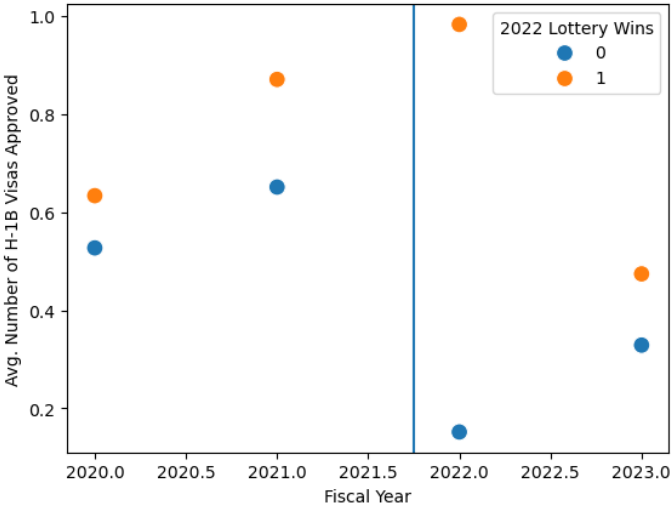
- **First stage:** number of H-1B visas granted (requires additional application after winning)
- **Hiring:** number of workers hired; number of total workers
- **Workers:** average promotion rate of new hires; turnover of incumbent workers

First Pass at First Stage

Subset to firms that only submit one application in a given lottery year to simplify visualization:

- Balanced sample of firms from preferred sample that are matched and can be consistently and uniquely identified in the H-1B data in all four years
- Subset to firms that submitted one application in 2022 (n = 1135)
- Plot mean number of H-1B visas approved

H-1B Visa Approvals by Lottery Win by Fiscal Year



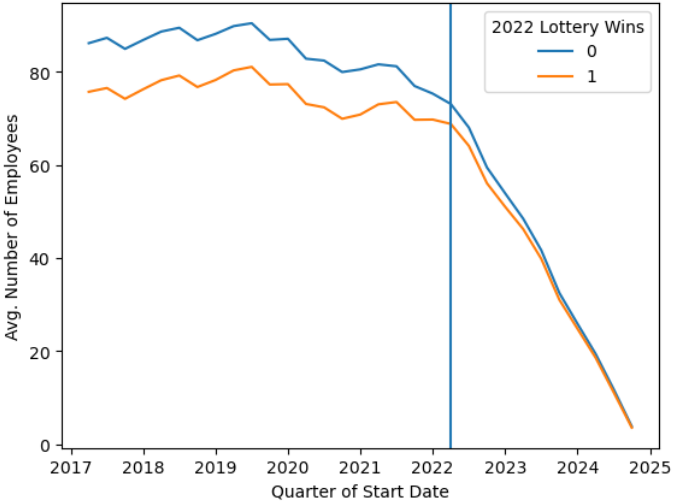
First Pass at Some Results

Subset to firms that only submit one application in a given lottery year to simplify visualization:

- Unbalanced sample of firms from preferred sample that are matched
- Subset to firms that submitted one application in 2022 (n = 13,758)
- Plot mean number of new hires and total employment in LinkedIn data

New Hires by Lottery Win by Quarter

Total Employment by Lottery Win by Quarter



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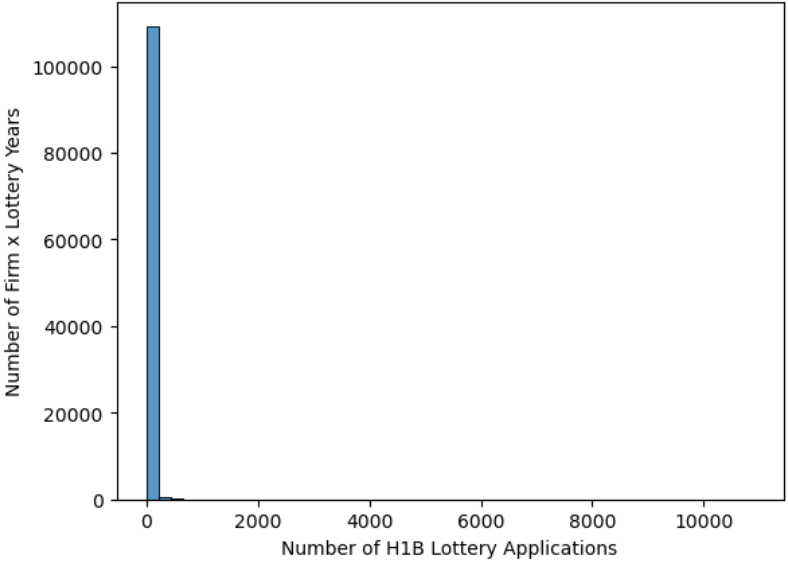
1. Look into merging on individual level
2. Run regressions on first stage and crowd-out
3. Construct measure of promotion/turnover

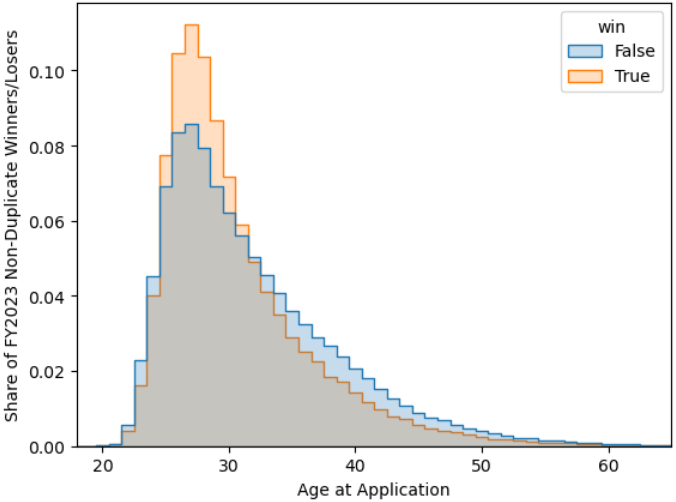
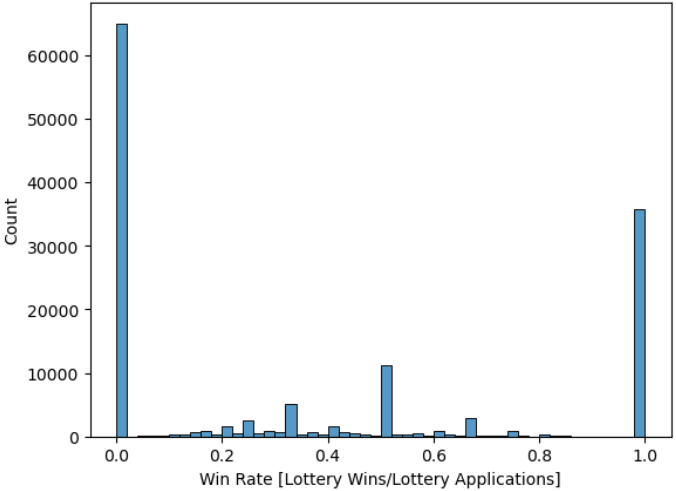
Raw Lottery App Dist. (Firm x Year Counts FY2021-2024)

back



Dist. of Age at Application (Preferred Sample FY2021-2024)

back



Dist. of Win Rates (Firm x Year Counts FY2021-2024) back